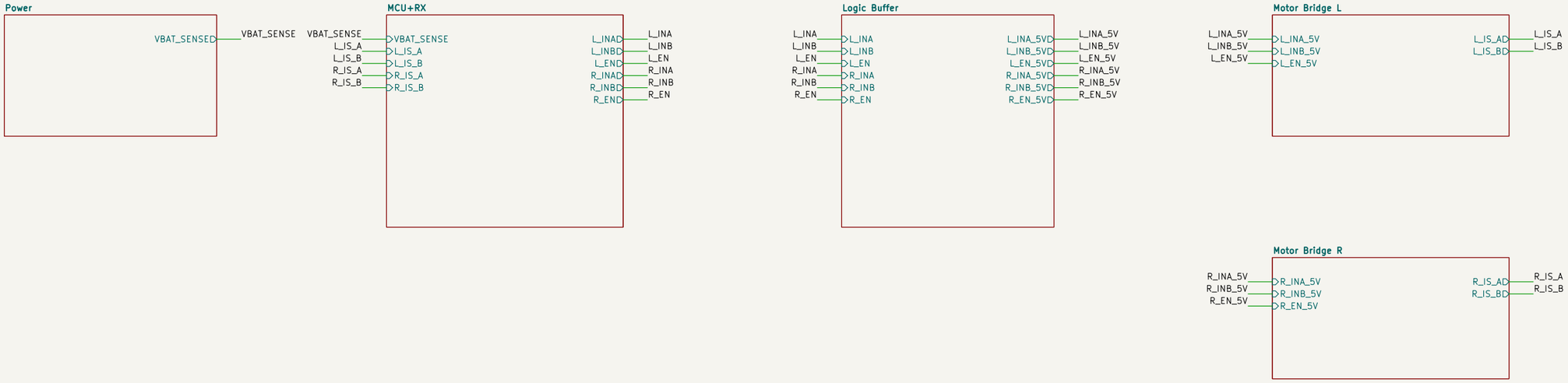
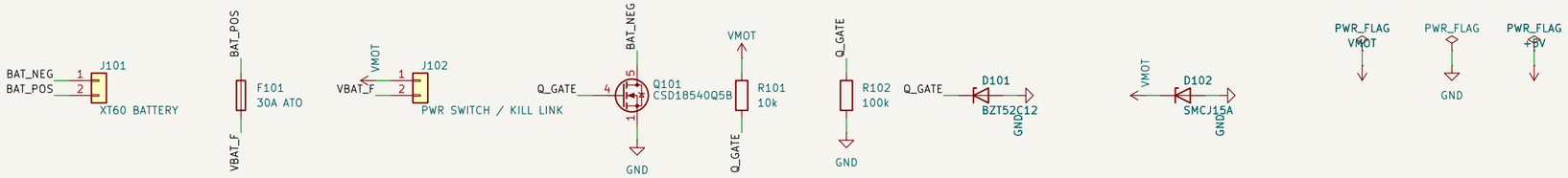


racebot_main – ESP32 + ELRS + 2x BTS7960B H-bridge main board (3S LiPo)
Power nets: VMOT (switched, fused battery), +5V (LM2596), +3V3 (ESP32 board LDO), GND.
Logic path: ESP32 GPIO (3.3 V) -> 74HCT244 (5 V) -> 10k -> BTS7960B IN/INH.
Current sense: BTS IS -> 470R -> 1k/10nF -> BAT54S clamp -> ESP32 ADC1.

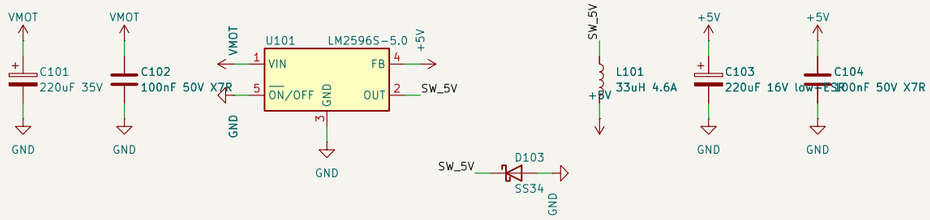
racebot

Sheet: /		
File: racebot_main.kicad_sch		
Title: racebot_main – Root		
Size: A3	Date: 2026-10-28	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/6

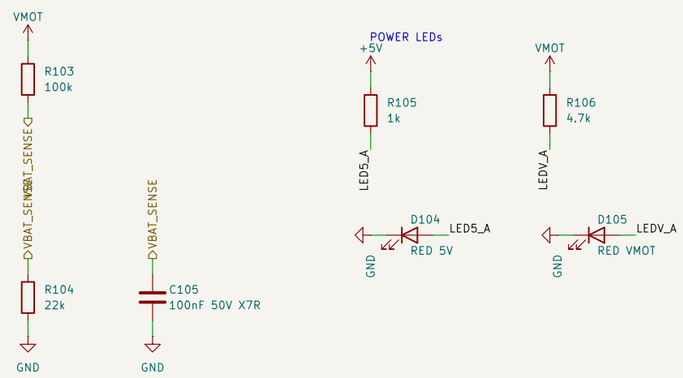
POWER INPUT / PROTECTION
XT60 (battery 3S, 9.0–12.6 V) -> 30 A ATO fuse -> VBAT_F -> J102 power switch / kill link -> VMOT.
Q101: low-side N-MOSFET reverse-polarity protection in the ground return (D=battery -, S=GND).
Gate fed from VMOT (after the switch) so nothing drains the battery when switched off.



5 V BUCK (LM2596S-5.0) -> ESP32 5V pin, ELRS receiver, 74HCT244
Keep VIN cap – U101 – D103 – C103 loop tight (see LAYOUT_NOTES.md).

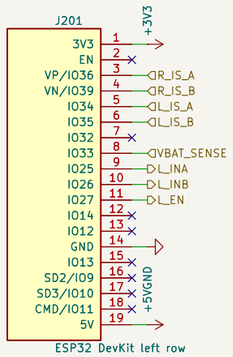


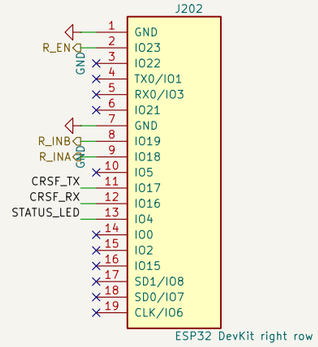
BATTERY SENSE: VMOT -> 100k/22k (2.27 V @ 12.6 V) -> GPIO33 (ADC1_CH5)



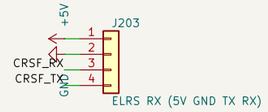
racebot
Sheet: /Power/
File: power.kicad_sch
Title: racebot_main – Power
Size: A3 | Date: 2026-10-28 | Rev: A
KiCad E.D.A. 10.0.5 | Id: 2/6

ESP32 38-pin DevKit (ESP-WROOM-32, DevKitC V4 pin order) on 2x 1x19 female headers.
Pin 1 of each row = antenna end. VERIFY against the board silkscreen before ordering.
Board is powered through its 5V pin from the LM2596. 3V3 pin = board LDO output (clamps only).

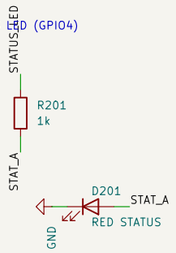




ELRS RECEIVER (SpeedyBee ELRS, CRSF @ 420k baud on UART2)
Receiver TX -> GPIO16 (ESP RX2), Receiver RX <- GPIO17 (ESP TX2)



STATUS LED (GPIO4)



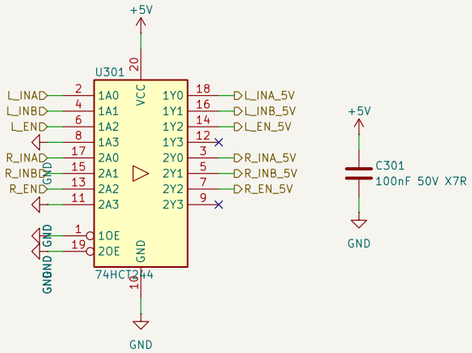
racebot

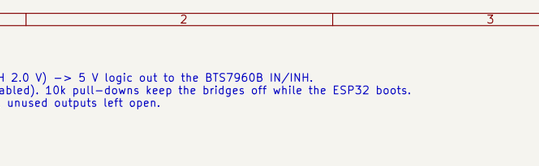
Sheet: /MCU+RX/
File: mcu_rx.kicad_sch

Title: racebot_main — MCU+RX

Size: A3	Date: 2026-10-28	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 3/6

74HCT244: 3.3 V ESP32 logic in (VIH 2.0 V) -> 5 V logic out to the BTS7960B IN/INH.
/1OE and /2OE tied low (always enabled). 10k pull-downs keep the bridges off while the ESP32 boots.
Unused inputs 1A3/2A3 tied to GND, unused outputs left open.

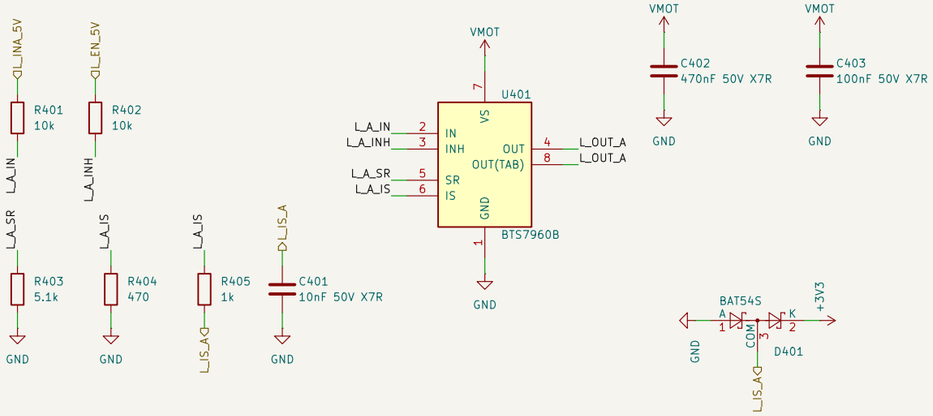




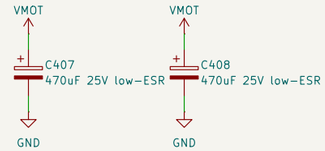
racebot		
Sheet: /Logic Buffer/ File: logic_buffer.kicad_sch		
Title: racebot_main – Logic Buffer		
Size: A3	Date: 2026-10-28	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 4/6

MOTOR BRIDGE L — 2x BTS7960B full H-bridge, left-side motors (2 in parallel). Plan ~20 A stall.
IS: 470R to GND (7 mA fault -> 3.3 V, ~1.1 V at 20 A with kILIS 8500), 1k + 10nF RC, BAT54S clamp to GND/3V3.
SR 5.1k sets slew rate (0603: swap 0R = fastest / 51k = slowest).

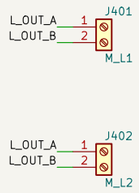
Half-bridge A



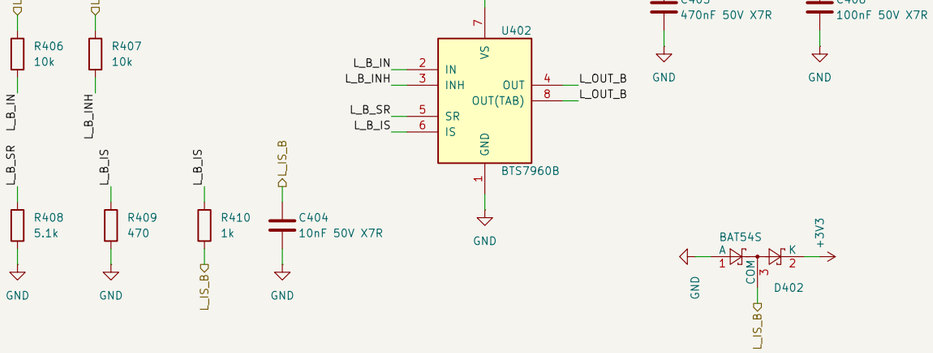
VMOT BULK (place at the bridge, star GND point)



MOTOR TERMINALS (2 motors in parallel)



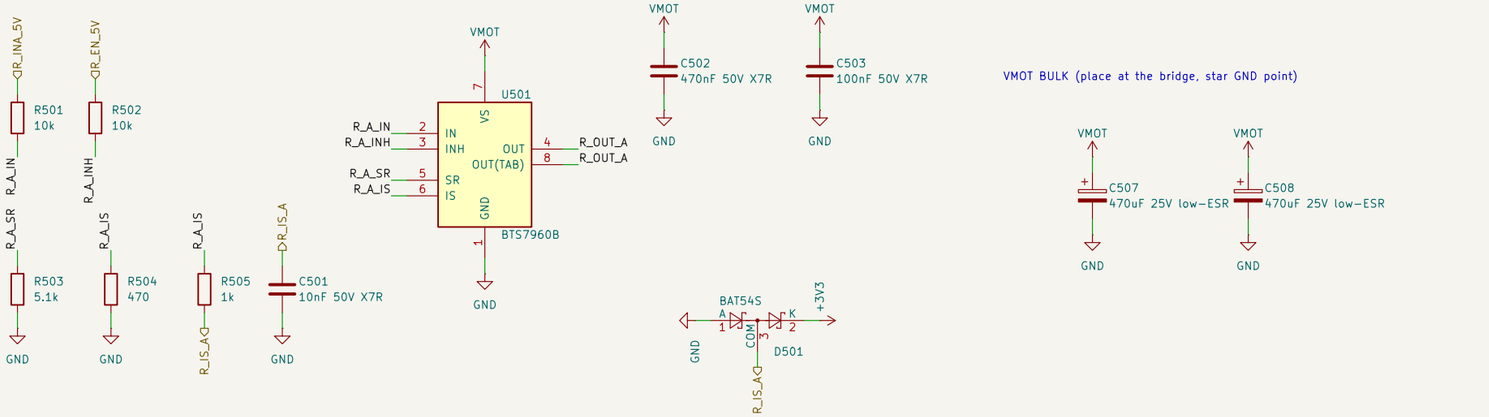
Half-bridge B



racebot
Sheet: /Motor Bridge L/
File: motor_bridge_L.kicad_sch
Title: racebot_main — Motor Bridge L
Size: A3 | Date: 2026-10-28 | Rev: A
KiCad E.D.A. 10.0.5 | Id: 5/6

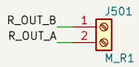
MOTOR BRIDGE R — 2x BTS7960B full H-bridge, right-side motors (2 in parallel). Plan ~20 A stall.
IS: 470R to GND (7 mA fault -> 3.3 V, ~1.1 V at 20 A with kILIS 8500), 1k + 10nF RC, BAT54S clamp to GND/3V3.
SR 5.1k sets slew rate (0603: swap 0R = fastest / 51k = slowest).

Half-bridge A

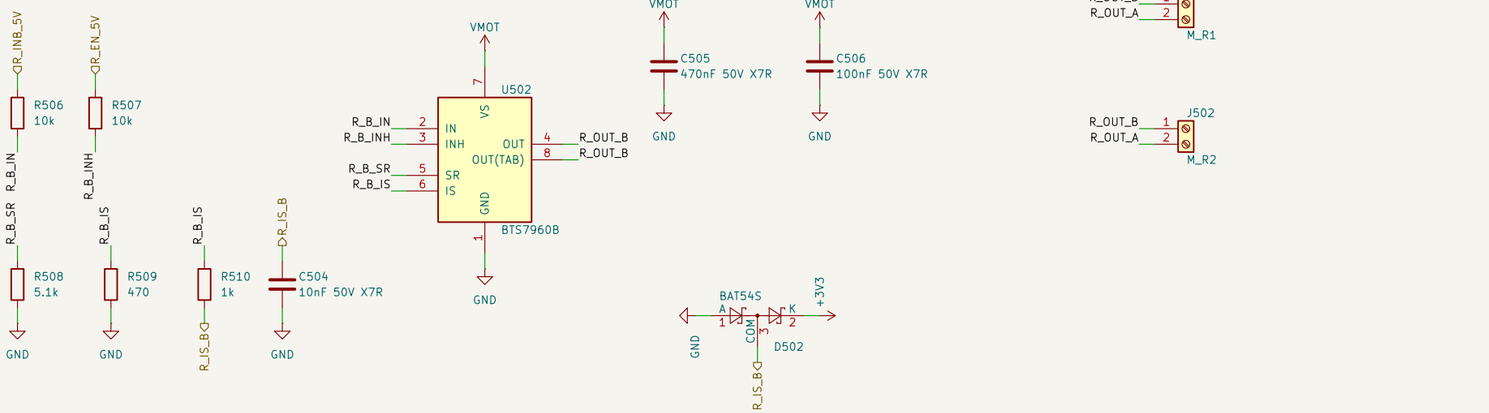


VMOT BULK (place at the bridge, star GND point)

MOTOR TERMINALS (2 motors in parallel)



Half-bridge B



racebot
Sheet: /Motor Bridge R/
File: motor_bridge_R.kicad_sch
Title: racebot_main — Motor Bridge R
Size: A3 | Date: 2026-10-28 | Rev: A
KiCad E.D.A. 10.0.5 | Id: 6/6